

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of B. Tech (EC/IT/CE) Examination

January 2016

EC 101 Basics of Electronics Engineering

Date: 02.01.2016, Saturday

Time: 01.30 p.m. To 04.30 p.m.

Maximum Marks: 70

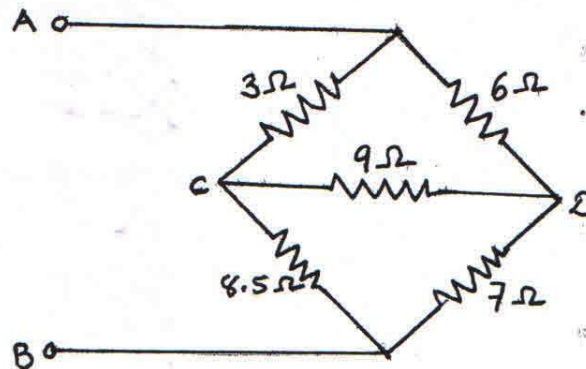
Instructions:

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

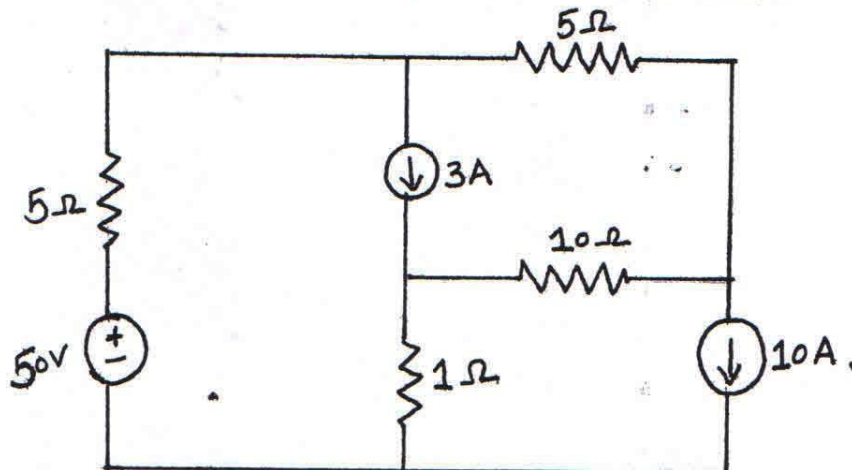
SECTION - I**Q - 1 Define the followings.****[07]**

- | | |
|------------|-------------|
| 1. Node | 2. Junction |
| 3. Loop | 4. Mesh |
| 5. Network | 6. Circuit |
| 7. Branch | |

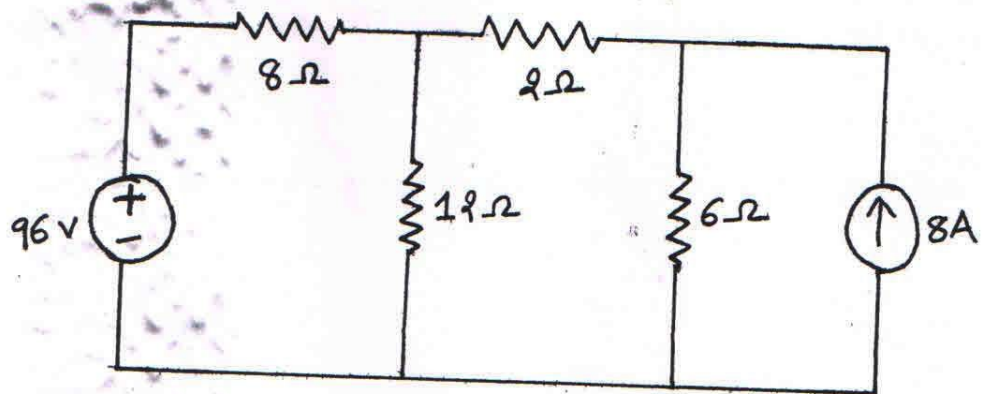
Q - 2.a Using Star Delta transformation, find the value of the equivalent resistance between points A and B of the circuit shown in figure. **[06]**

**Q - 2.b Answer any two questions.****[10]**

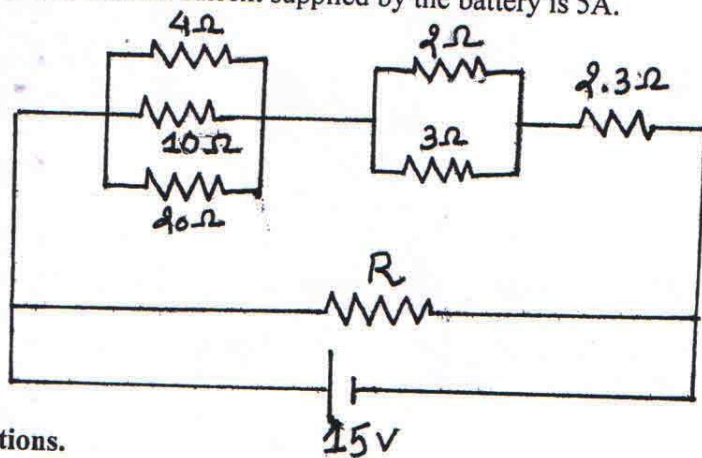
- (i) Find current in the 10Ω Resistor for the given circuit using mesh analysis.



- (ii) Determine the voltage across 6Ω Resistor for the following figure using node analysis.



- (iii) Determine the value of R so that the current supplied by the battery is 5A.



Q - 3 Answer any two questions.

- Convert decimal number 250.5 to base 3, base 4, base 7, base 8 and base 16.
- Implement following using basic gates.
 - $XY + XY'$
 - $(X + Y)(X + Y')$
 - $XYZ + X'Y + XYZ'$
 - $ABC + A'B'C' + A'BC + ABC' + A'B'C$
- With neat and clean diagram, explain Full-wave rectifier and derive its equation for rectification efficiency.

[12]

SECTION - II

Q - 4 Define the followings.

- | | |
|-------------------------|--------------------|
| 1. Breakdown voltage | 2. Leakage Current |
| 3. Ripple factor | 4. Knee voltage |
| 5. Peak Inverse Voltage | 6. Doping |
| 7. Time Constant | |

[07]

Q - 5.a Two slabs of material of dielectric constants 4 and 6 having thickness of 2mm and [07]

5mm respectively are inserted between plates of a parallel plate air capacitor. Find by how much distance between the plates should be changed so as to restore the original value of capacitance.

OR

Q – 5.a A $2\mu\text{F}$ capacitor is connected in series with $1\text{M}\Omega$ resistor across a 100V d.c. supply. [07]
Calculate

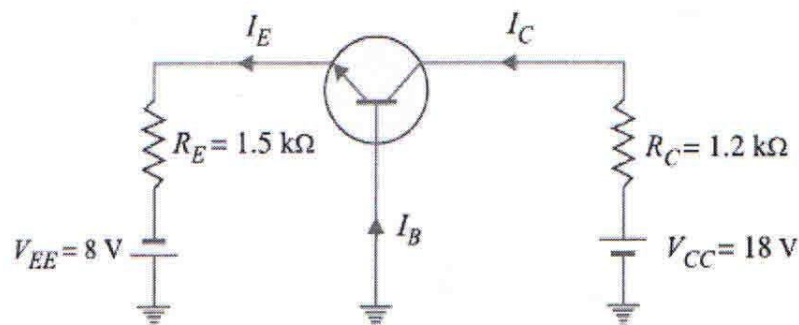
1. Time constant
2. Initial charging time
3. Voltage across capacitor 6seconds after the switch has been closed
4. Time taken for the capacitor to be fully charged
5. Time taken for the capacitor voltage to reach 75 volts.

Q – 5.b Answer any two questions. [08]

- 1) Define Electric Field, Electric flux Density and Electric field Intensity and derive the relationship between them.
- 2) Derive the equations of E.M.F. for transformer.
- 3) Describe the action of the following filter circuits : (1) Capacitor filter (2) π - filter

Q -- 6.a Explain charging of a capacitor and derive the equation for charging voltage across a capacitor. [07]

Q -- 6.b For common base circuit shown in Figure. Determine I_C and V_{CB} . Assume transistor to be of silicon. [06]



OR

Q – 6.a Derive an equation for energy stored in a capacitor. [06]

Q – 6.b Explain input and output characteristics of CE configuration, also derive the relationship between α and β . [07]
